Input system:
Dependent variables:y, x
Independent variables:t
Equations:

$$\frac{dx}{dt} = ax - bxy\text{inv}\left(\left(1 + Ax\right)\left(1 + By\right)\right)$$

$$\frac{dy}{dt} = \left(-1\right)cy + dxy\text{inv}\left(\left(1 + Ax\right)\left(1 + By\right)\right)$$

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Result:
Dependent variables:y, s_1, x
Independent variables:t
Additional variables:

$$s_1 = \text{inv}\left(\left(1 + Ax\right)\left(1 + By\right)\right)$$

Equations:

$$\frac{dx}{dt} = \left(-1\right)bxys_1 + ax$$

$$\frac{dy}{dt} = dxys_1 + \left(-1\right)cy$$

$$\begin{aligned}\frac{ds_1}{dt} = {} & s_1^3By^2Abx + \left(-1\right)s_1^3Ax^2Bdy + s_1^3Abxy + \left(-1\right)s_1^2ByAax \\ & + \left(-1\right)s_1^3Bdxy + s_1^2AxBcy + \left(-1\right)s_1^2Aax + s_1^2Bcy\end{aligned}$$